Analysis of CFD methodologies applied to a high-pressure H_2 injector in support to the design process

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RELATORI
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Ing. Giuseppe Cicalese
Ing. Veronica Patrizi
Ing. Sara Vongher

DUMAREY

R&D COMPUTATIONAL FLUID DYNAMICS

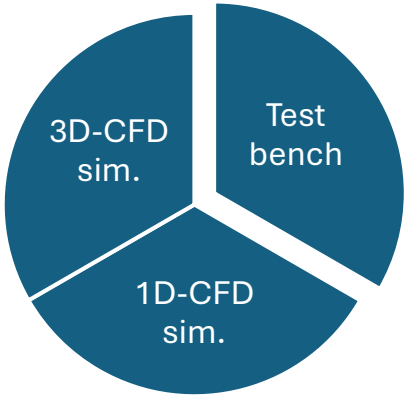
AGENDA

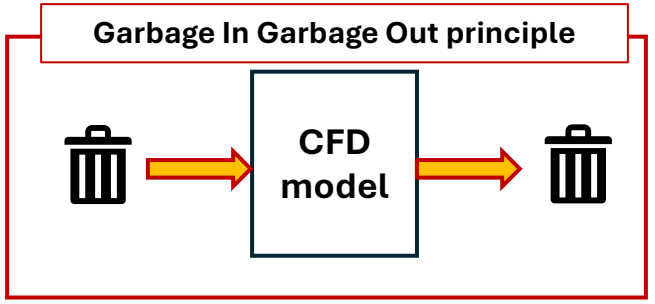
1. INTRODUCTION
2. MODELS SETUP
3. STEADY – STATE APPROACH
4. DFBI – OVERSET 2D APPROACH
5. CONCLUSIONS AND FINAL CONSIDERATIONS

Design process  **Computational Fluid Dynamic**(CFD)

CFD properties in the design process

- ✓ **Speed**: faster insights of many different configurations in hours
- ✓ **Cost-effective**: cheaper than multiple experimental tests
- ✓ **Detailed analysis:** easy access to specific quantitative data and analysis of complex phenomena
- ✓ **Problem-solving:** Helps identify and fix design flaws



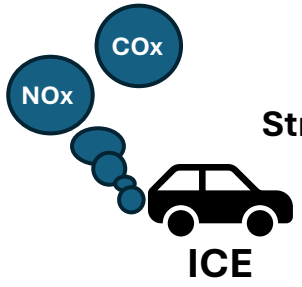


⚠ Accurate but may be misleading

❖ Numerical correlation with Exp. Data

Experimental measurements:

- **Tune** / calibrate the **model**
- **Check** model **reliability**



Straightforward transition





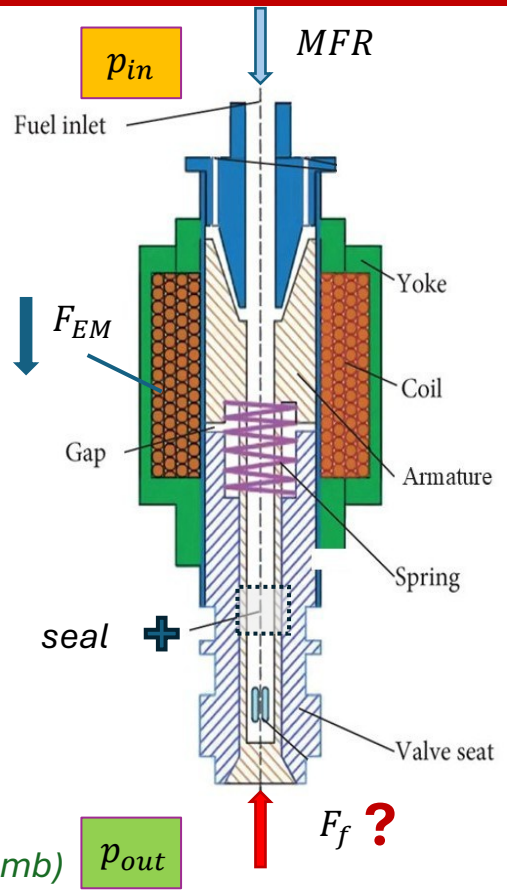
- flexible energy carrier
- NO carbon based emissions
- Reduction of rare material usage
- Established know-how
- Established manufacturing facilities and processes

⚠ Different fuel properties

NEW

- **New design challenges**
- **Different models**

INTRODUCTION

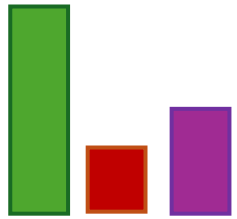


Main outcomes of interest:

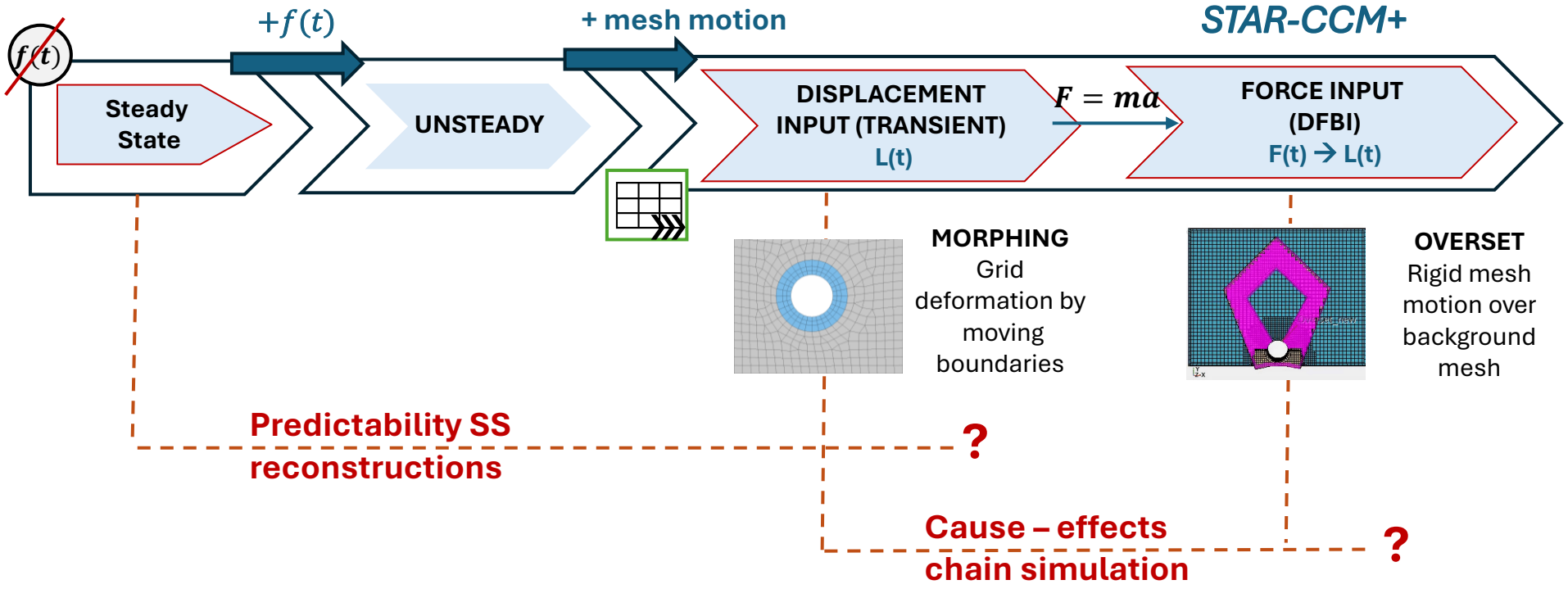
- Mass Flow Rate ($\boldsymbol{MFR}$)
- Axial fluid forces ($\boldsymbol{F_f}$)

GOAL

*Analyse and identify the **CFD methodology** that **provides the best support** for the different **stages of the virtual design process** of H2 injector.*



best trade-off between **accuracy**, **computational cost**, and **model complexity**



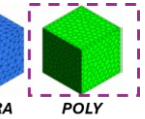
2 MODELS SETUP

MODEL SETUP

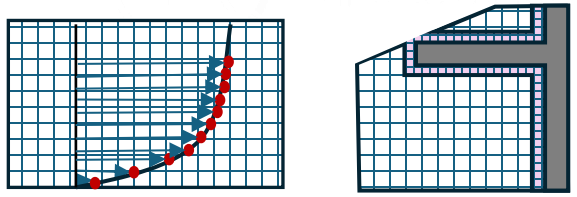
STAR-CCM+ CFD WORKFLOW

CAD .step

GEOMETRY PREPARATION



Generate computational grid



MESHING

CFD region

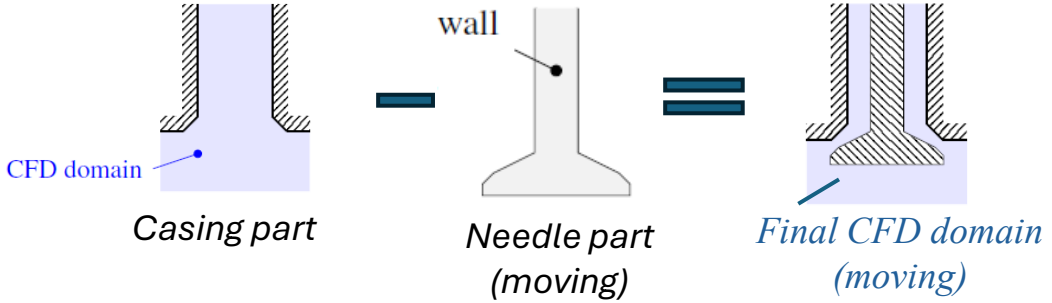
numerical domain

PHYSICS AND SOLVER

numerical system

VALIDATION

METHODOLOGY ASSESMENT



Boundary conditions, physics and solver parameters:

- Experimental operating conditions (p_{in}, p_{out})
- Compressible real single/multi-component gas (H_2 / H_2 -air)
- Time dependency and motion
- Governing equation solution algorithms (segregated/coupled)

Model reliability and results analysis

- Comparison with EXP-DATA (MFR)
- Numerical correlation between models (F_f)

Efficiency evalutation:

- Accuracy
- Computational cost
- Ease of set up (time and complexity)
- Design suitability

3 STEADY STATE APPROACH

STEADY STATE APPROACH

STEADY STATE: $f = f(x, y, z, \cancel{t})$

DYNAMIC
injection cycle → **SEQUENCE of *indipendent* steady state *operating conditions***

TRANS. MORPH. model

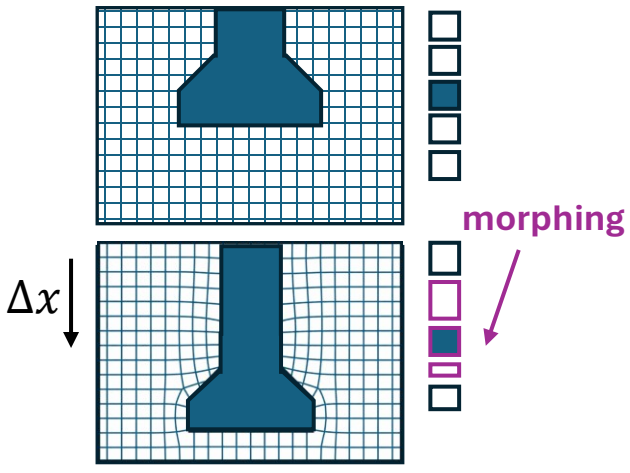
Moving geometry
(parts movement)

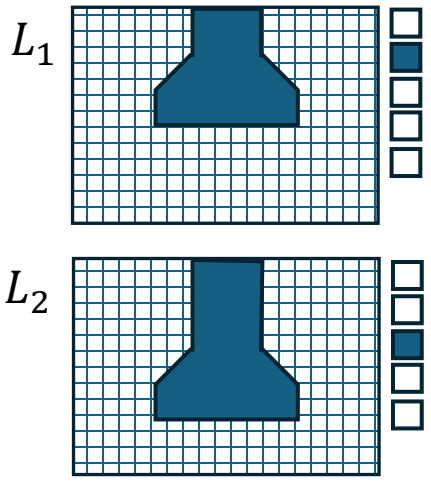
Flow evolution (t)
(time history effects)

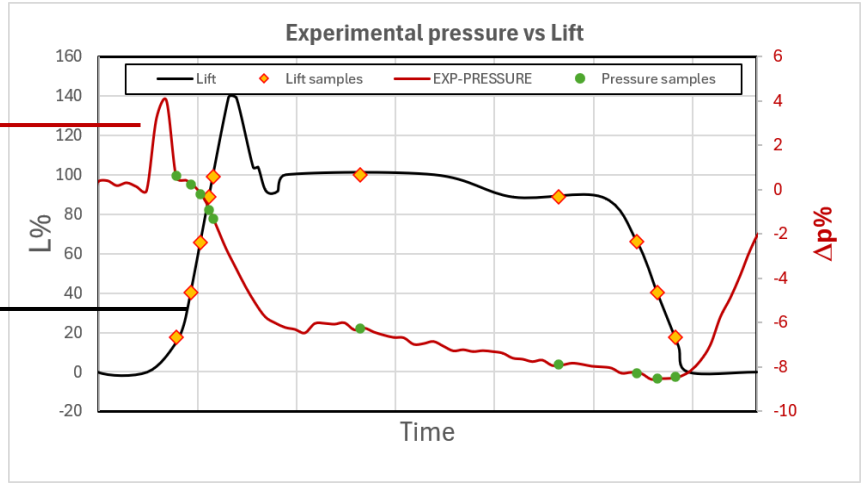
Steady-state assumptions

Freezed geometry
(no relative motion)

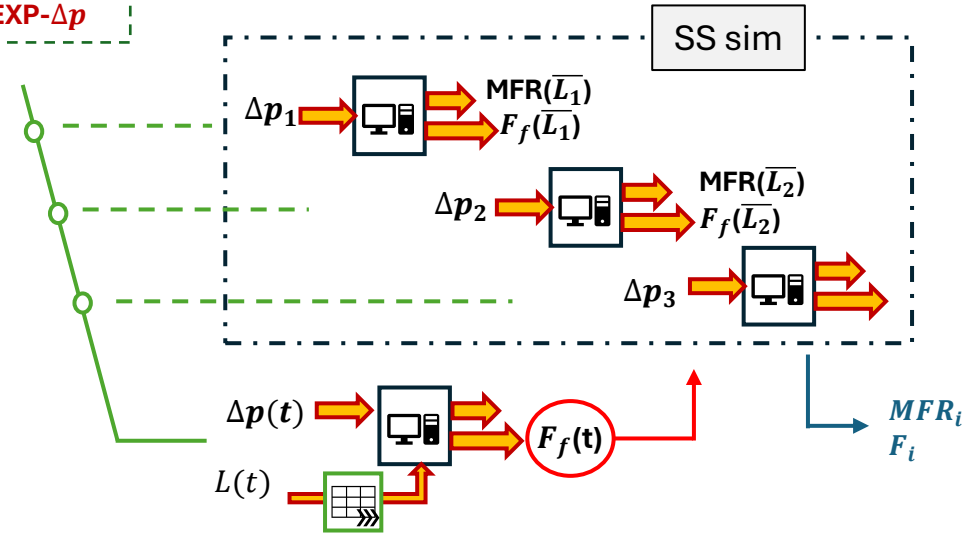
Fully developed flow
(no transient effects)







EXP-Lift
EXP-Δp



STEADY STATE APPROACH

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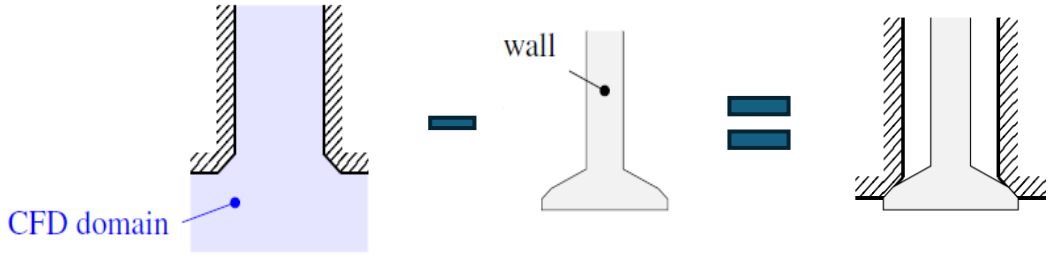
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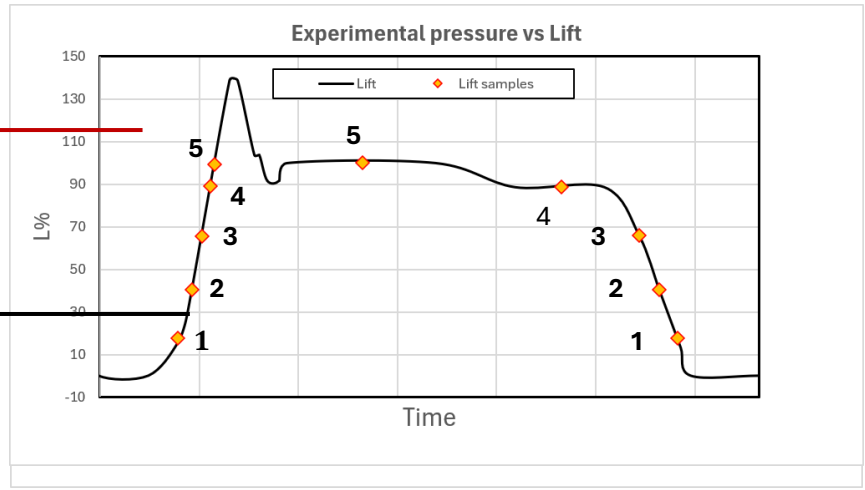
GEOMETRY PREPARATION

Additional pre-processing opration:

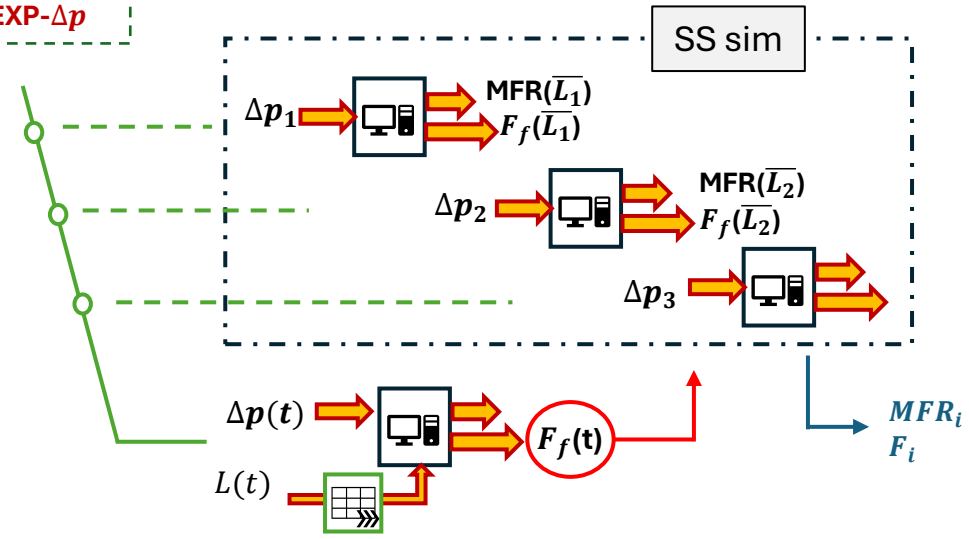
1. Needle translation L_i
2. Subtract operation for each congifiguration

◆ **x1 sim. each**





EXP-Lift
EXP-Δp



STEADY STATE APPROACH

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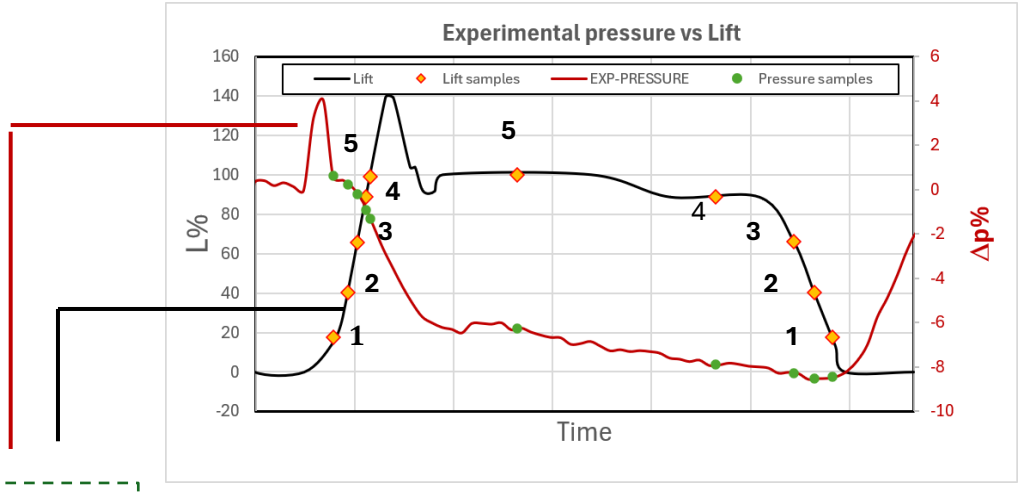
Flow evolution (t)
(time history effects)

Steady-state assumptions

Freezed geometry
(no relative motion)

Fully developed flow
(no transient effects)

p_5



EXP-Lift
EXP-Δp

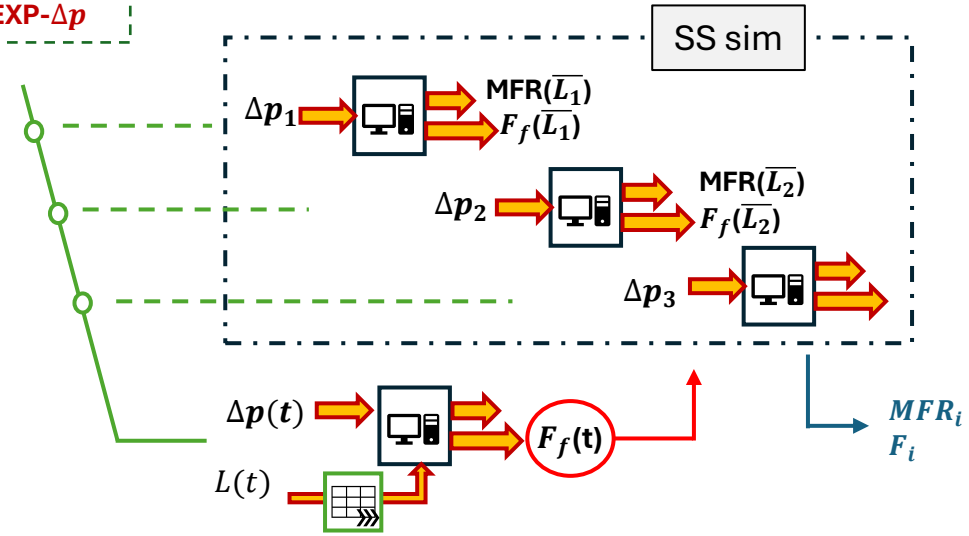
	P				
L [%]	p5	p4	p3	p2	p1
18					
40					
65					
89					
100					
100					
89					
65					
40					
18					

opening
hold
closure

	P				
L [%]	p5	p4	p3	p2	p1
18					
40					
65					
89					
100					

$F = f(p)$ **Opening vs Closure**

$F = f(L)$, **Motion effect**



STEADY STATE APPROACH

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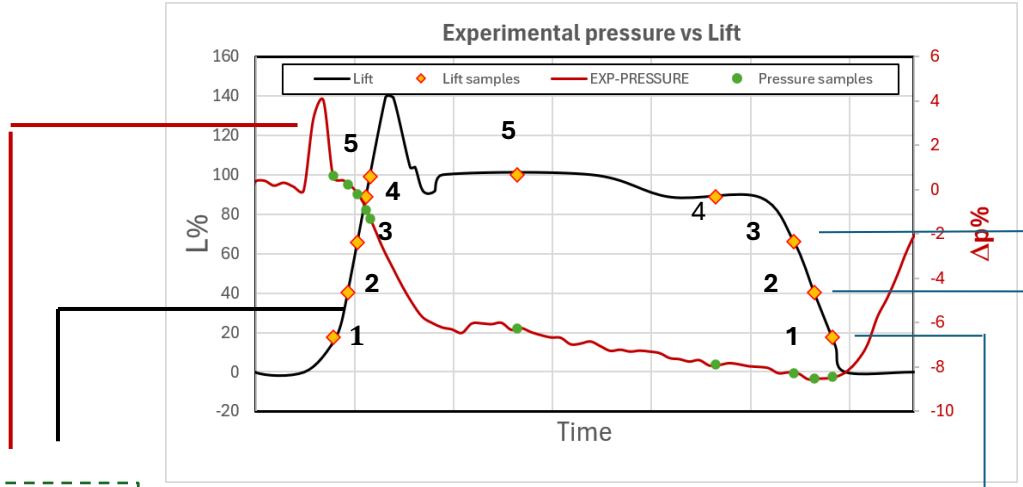
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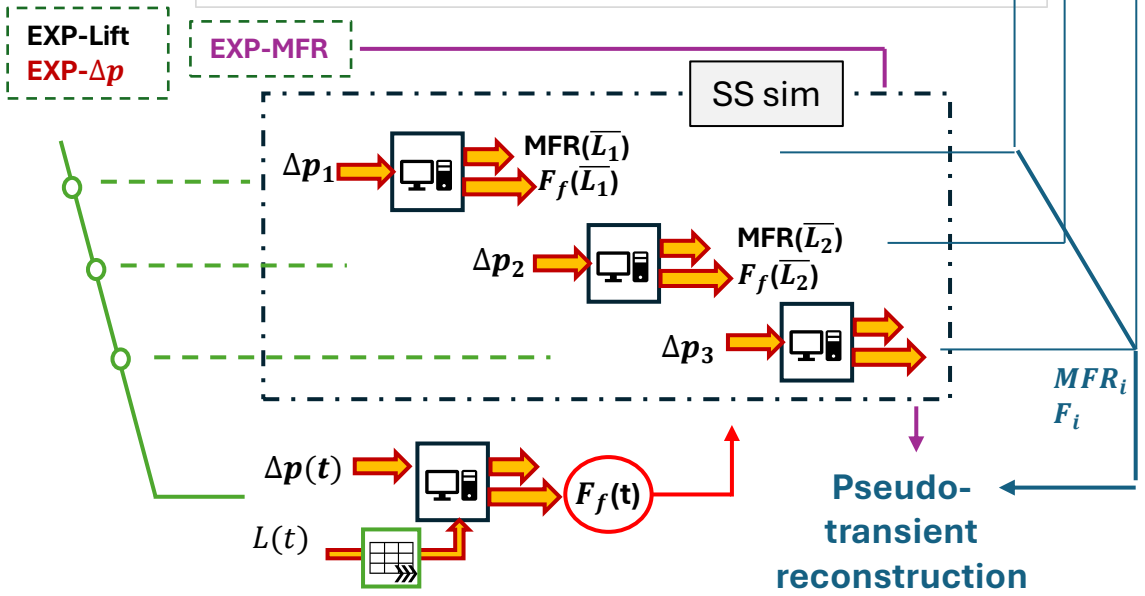
Flow evolution (t)
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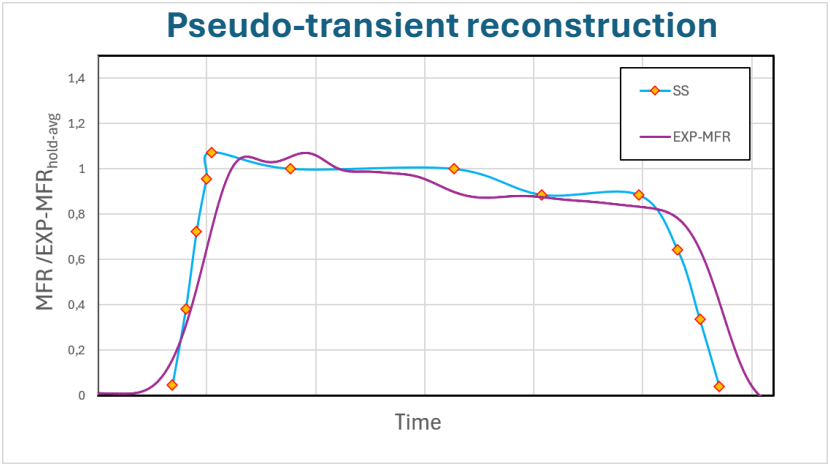
Steady-state assumptions

Freezed geometry
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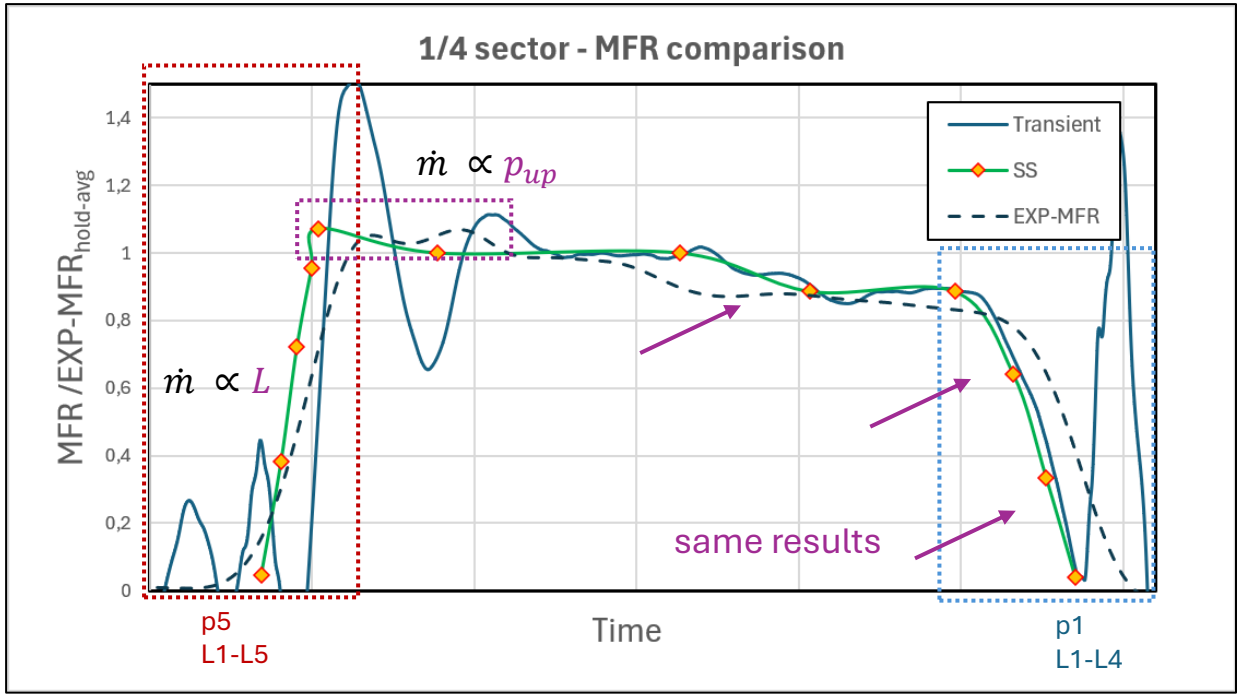
Fully developed flow
(no transient effects)







STEADY STATE APPROACH

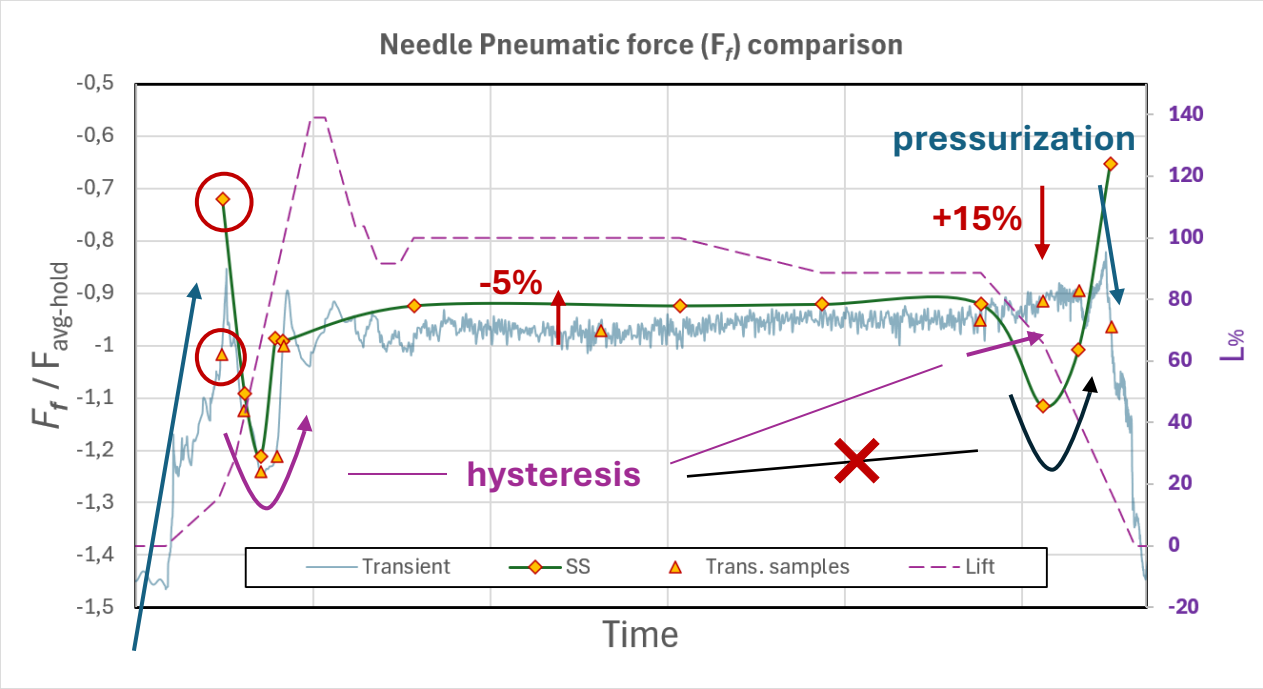


Steady-state **accurately predicts** injector performance **at a fraction of the computational cost**

MFR validation (EXP-comparison)

	SS	TRANSIENT
EXP. CORREL.	★★★	★★★
ACCURACY	★★★	★★☆
Opening	★★★	★★☆
Holding	★★★	★★☆
Closing	★★★	★★½
COST	↑	↑↑↑

- Excellent agreement with experimental MFR
- Expected physical dependency reproduced $\dot{m} \propto A \cdot P_{up}$
- Better or same average prediction than transient morphing



Steady-state **accurately predicts** injector performance **at a fraction of the computational cost**

..but **higher sensitivity** of pressure based parameters **to transient effects**

Fluid Force validation (Numerical Correlation)

SS comparison to **TRANSIENT** numerical reference

		Flow motion	Needle motion
Opening	★ ★ ⯪	→	→
Holding	★ ★ ★	→	/
Closing	⯪ ☆ ☆	→	←

(path dependence)

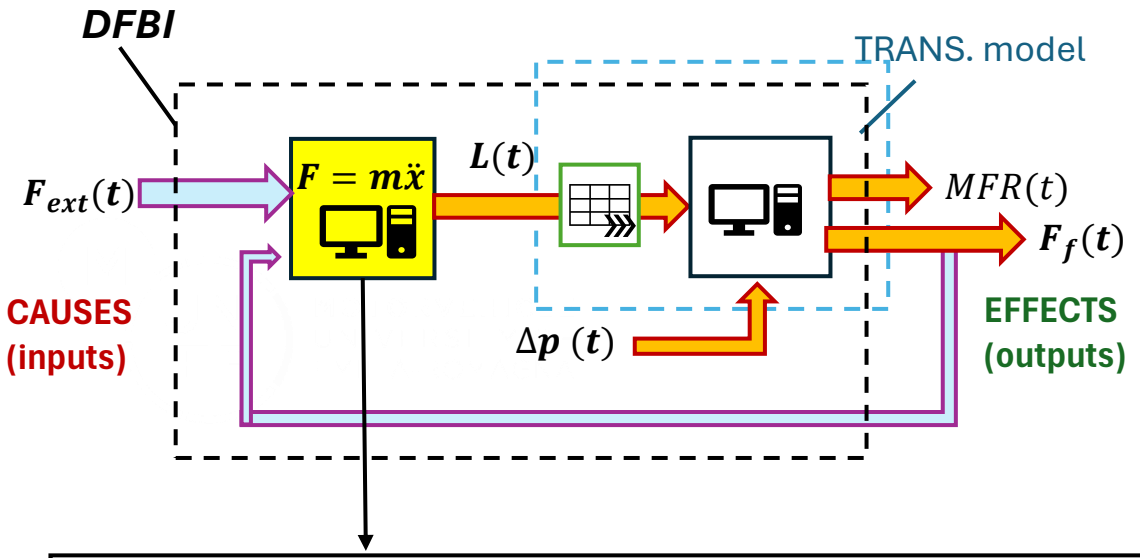
No hysteric behaviour detected

- ✓ **Remarkable overall level of agreement** on net closing effect
- ✓ **SS reliable as long as needle does not invert motion**
  - **Largest error** for positions characterized by **strong accelerations and transient phenoman**
- ✗ **No informations about sealing conditions and pressurization effects**

→ **Rapid design evaluation and output extrapolation**

4 DFBI – OVERSET METHODOLOGY

DFBI-OVERSET APPROACH



Dynamic Fluid Body Interaction $F_{ext}(t) \rightarrow L(t) \rightarrow F_f$

Motion **PRESCRIBED** ⟶ Motion **PREDICTED**

Further level of physical modelling

TRANS. model	**DFBI**
Imposed motion (simulation input)	**Motion computed** (part of solution)
X F_f **only effect**	F_f = **EFFECT and CAUSE** of dynamics
X L(t) primary input (no F_i)	$F_{ext}(t)$ = primary input

BODY entity

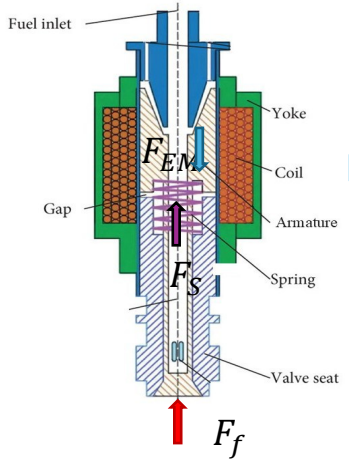
a. **Inertial properties** (mass)
b. **Parts**: needle boundaries
c. **kinematic constraints** → **translating motion**
d. **External forces**

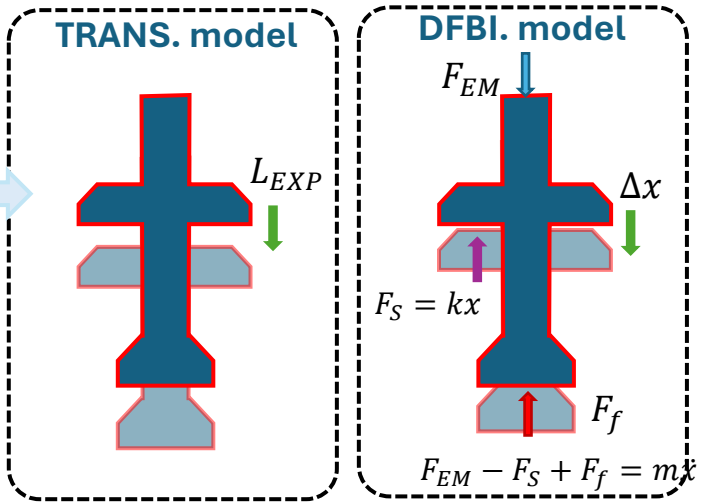
$$m\ddot{x} = ...$$

$-F_{EM}$
$F_0 + kx$
$k\ \max(0, -x^*)$
F_f

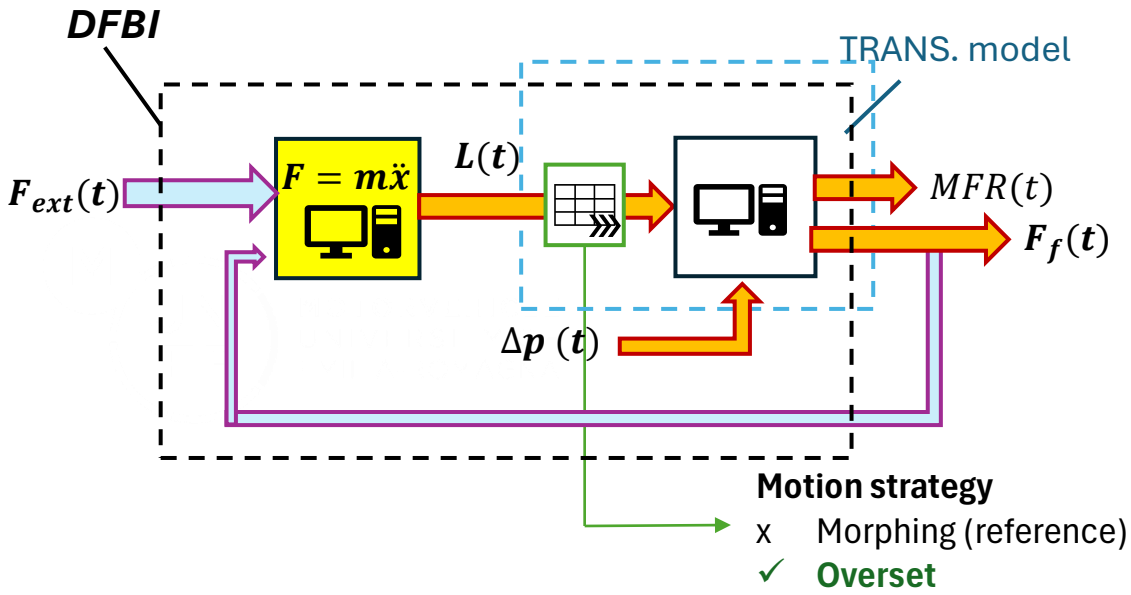
Needle part

- Magnetic Force (tabulated)
- Return spring
- Seal
- Fluid force (report)





DFBI-OVERSET APPROACH

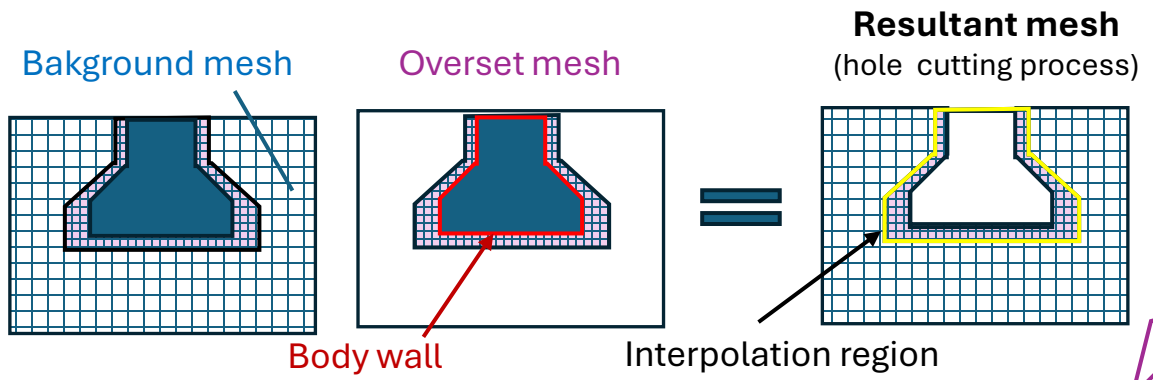


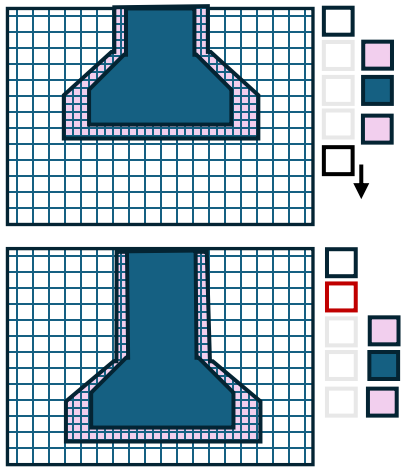
Motion strategy

x Morphing (reference)

✓ **Overset**

OVERSET: simplified motion technique employing **independent optimized meshes, overlapping** each other **over background region** on which the **solution** is **interpolated**





OVERSET

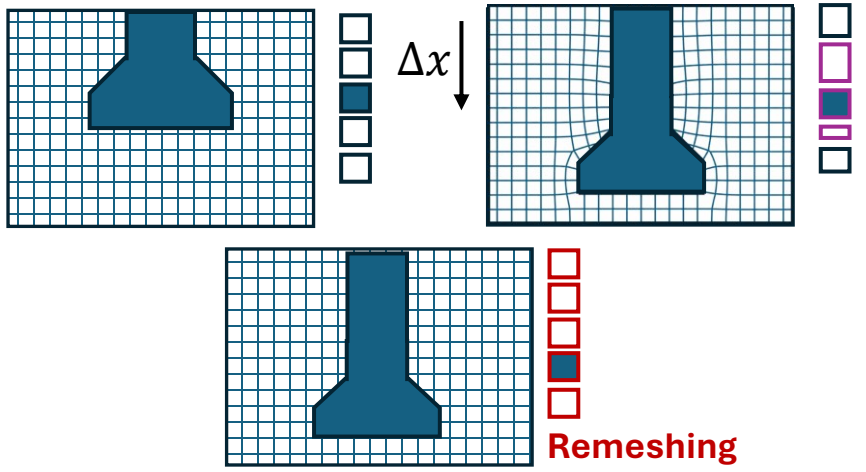
Coexisting meshes

↓ overlapping

Background cells **deactivation**

↓ **Overset motion**

Background activation



MORPHING

Initial mesh

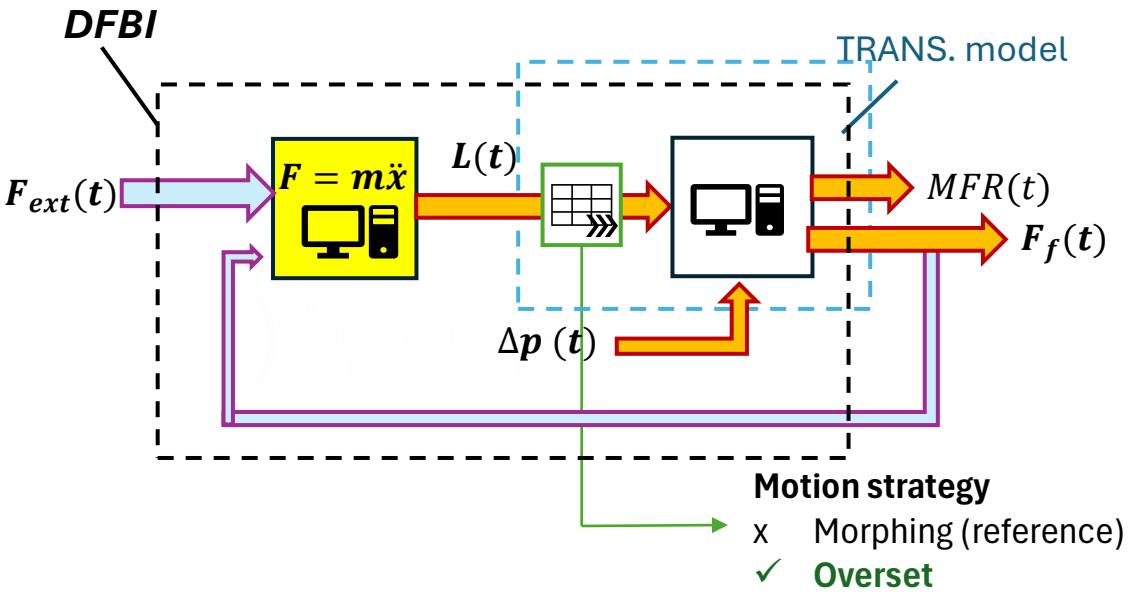
↓ Imposed displacement

Mesh deformation

↓ **Mesh quality ↓**

Remeshing (New undeformed mesh)

DFBI-OVERSET METHODOLOGY

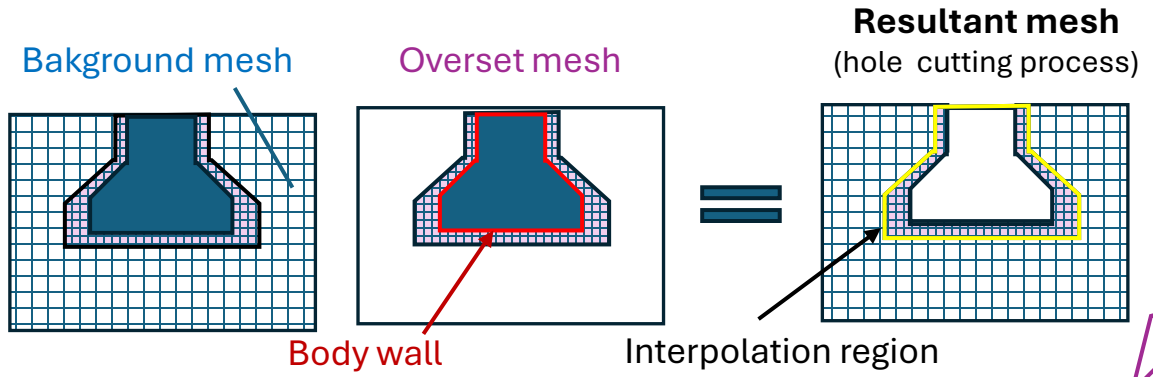


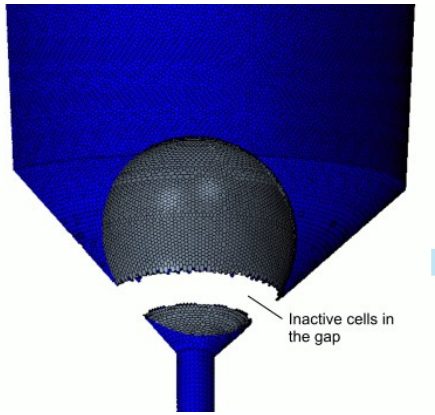
Motion strategy

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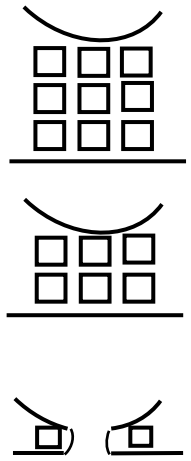
✓ **Overset**

OVERSET: simplified motion technique employing **independent optimized meshes, overlapping** each other **over background region** on which the **solution** is **interpolated**





Gap Closure model (contacts)



MORPHING ✗

1 mesh

Surface moves → Gap → 0

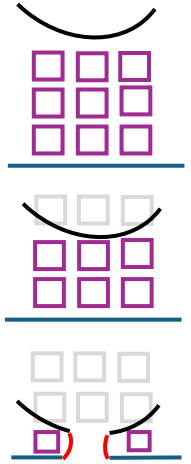
↓ $\Delta z < \bar{z}$

Gap closure interface → same part surface

↓ **Surfaces merge**

Motion bloked (even if $F_{ext} \neq 0$)

CONTACT ✗



OVERSET ✓

2 bodies → 2 meshes (background, overset)

Body moves → **Rigid mesh TRANSLATION**

↓ $\Delta z < \bar{z}$

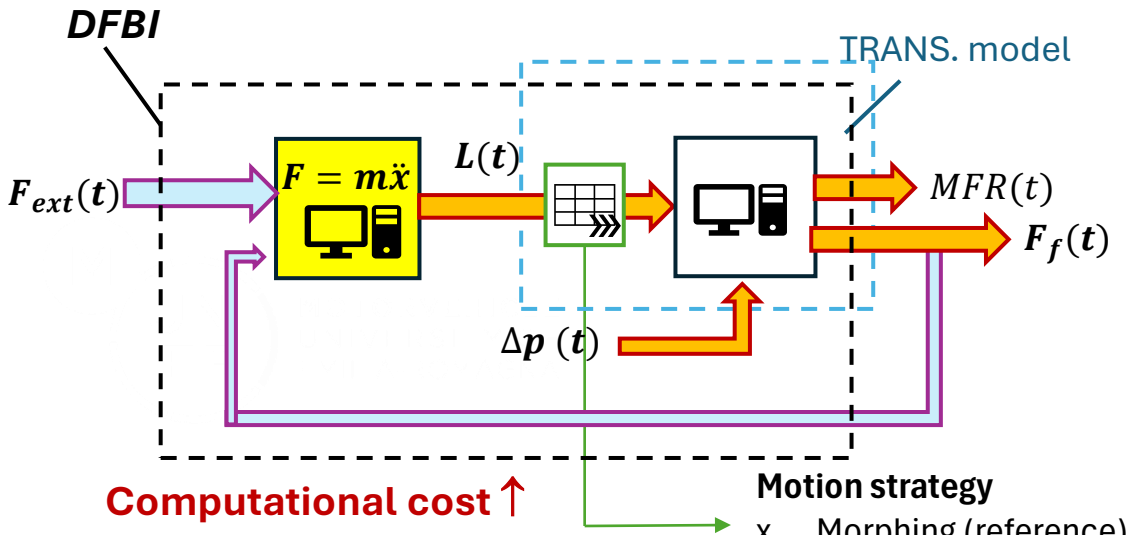
Gap → 0 separated surfaces closed

↓ **Cell deactivated**

Motion preserved

CONTACT ✓

DFBI-OVERSET APPROACH



Computational cost ↑

Motion strategy

x Morphing (reference)

✓ **Overset**

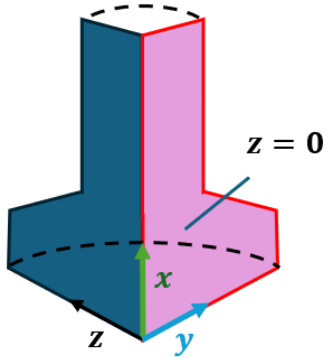
Limited resources

Because of:

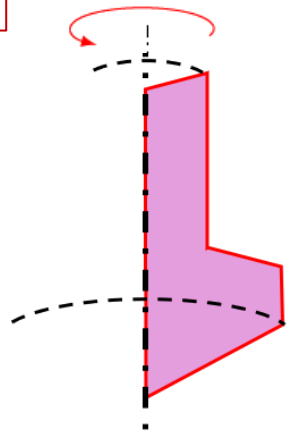
- Time-dependency
- Fluid-Structure inter.
- Multiple mesh handling

2D approximation to save computational power

2D-axialsym. model approximation



3D geom. (quarter)



2D representative section

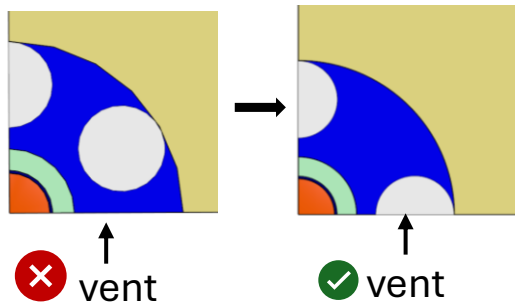
Requirement: **1 single section** to be **representative** of the **whole fluid domain** through **axialsymmetric revolution**

Main problem → **hole patterns** with **different angular arrangement**; always pairwise alignment within eligible cutting plane

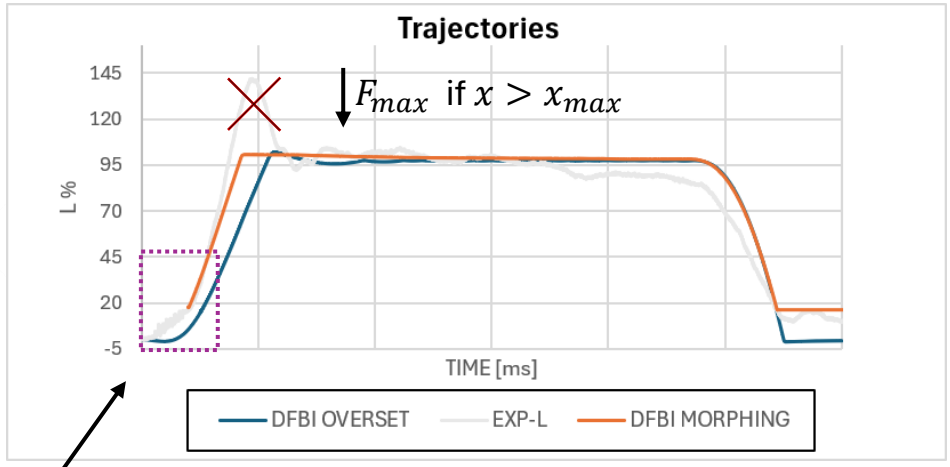
disconneted flow passages ⚠

A. **Plane yz** 1 2 ✓ 3

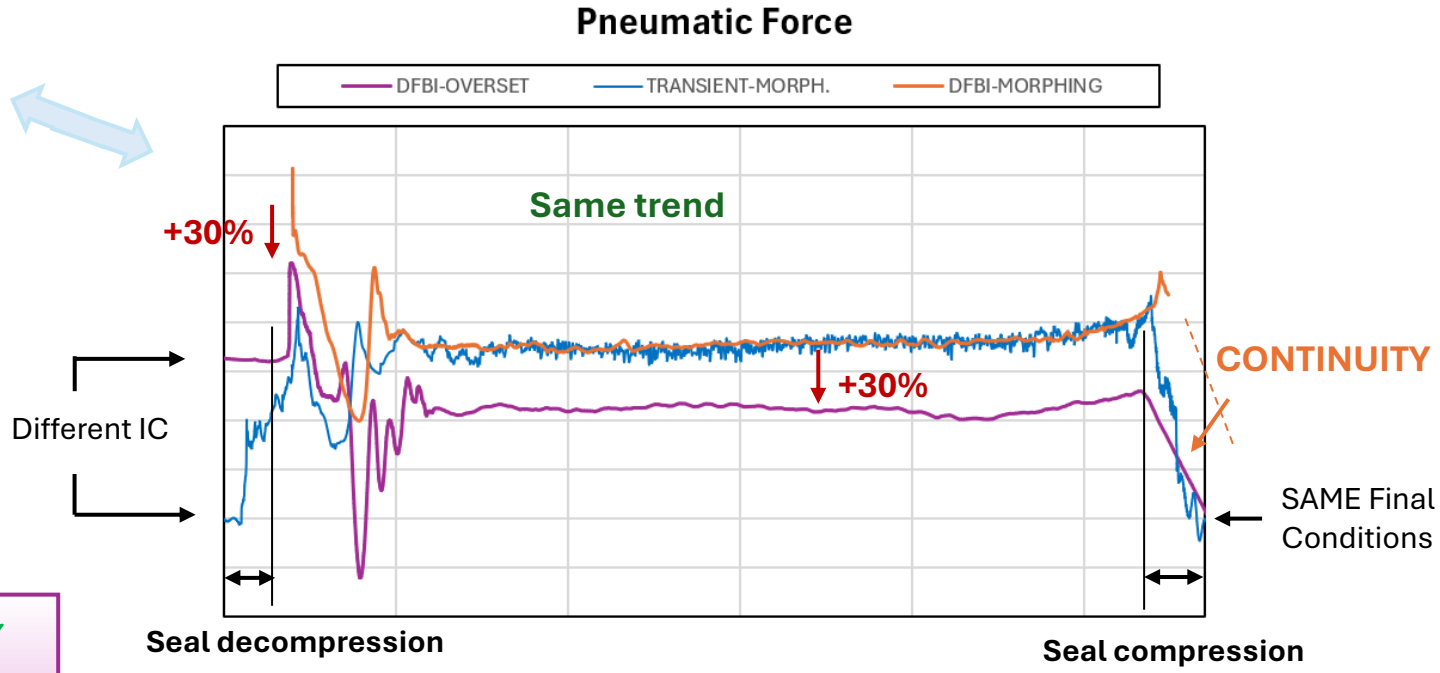
B. **Plane xz** 3 2 ✓ 1



DFBI-OVERSET APPROACH



- **Morphing**: extrapolated initial opened condition
- **Overset: delayed and slower** (free evolution)



DFBI: Nominal needle **motion predicted from forces only** ✓
OVERSET: sealing **contacts** correctly **reproduced**

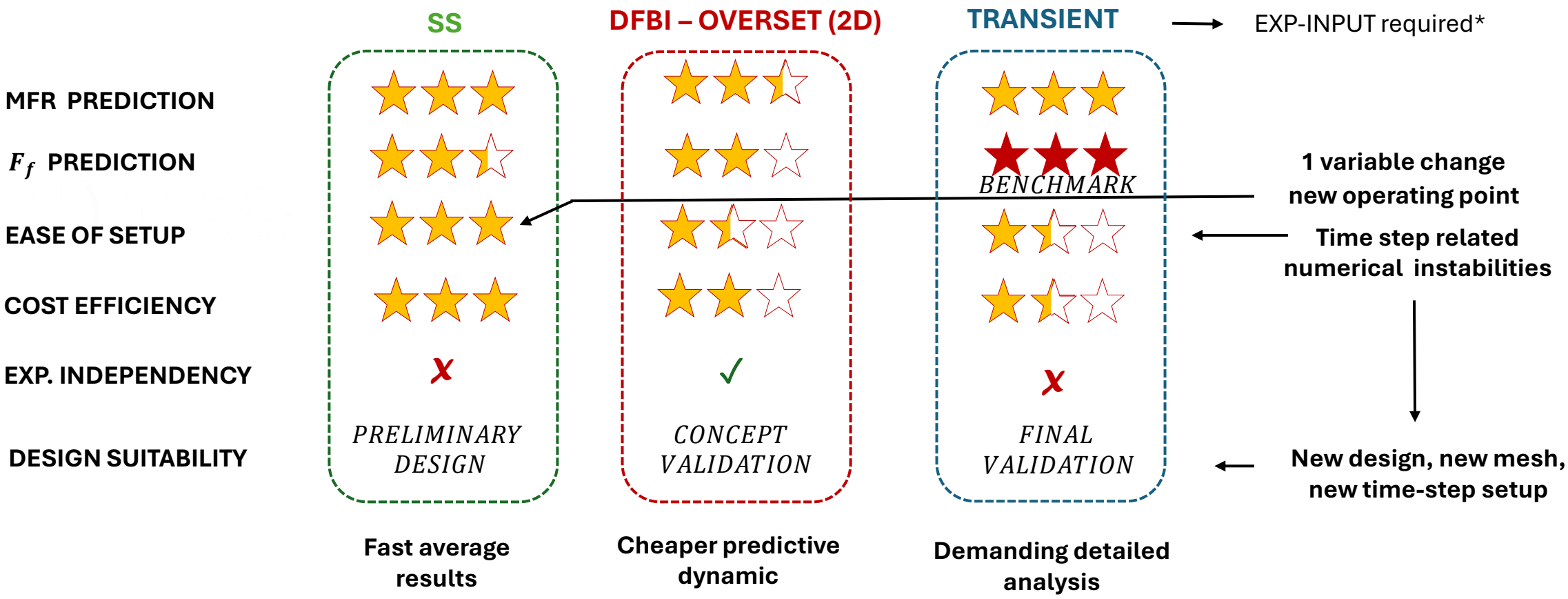
No overshoots/undershoots → finer needle dynamic and enviroment interaction required

Approximated reliable predictive tool for concepts validations

	TREND	ACCURACY	SEALING	COST
DFBI 2D-OVERSET	★★★	★★☆	✓	> -50%
DFBI MORPHING	★★★	★★★	✗	+20%
TRANSIENT	*BENCHMARK*		✓	*BENCHMARK*

4 FINAL CONSIDERATIONS

FINAL CONSIDERATIONS



The right CFD model is **not** necessarily the **most complex one**, **but the one** that **provides the required information** with the lowest engineering effort in the **right stages** of the desing development.

THANK YOU FOR YOUR ATTENTION

QUESTIONS?